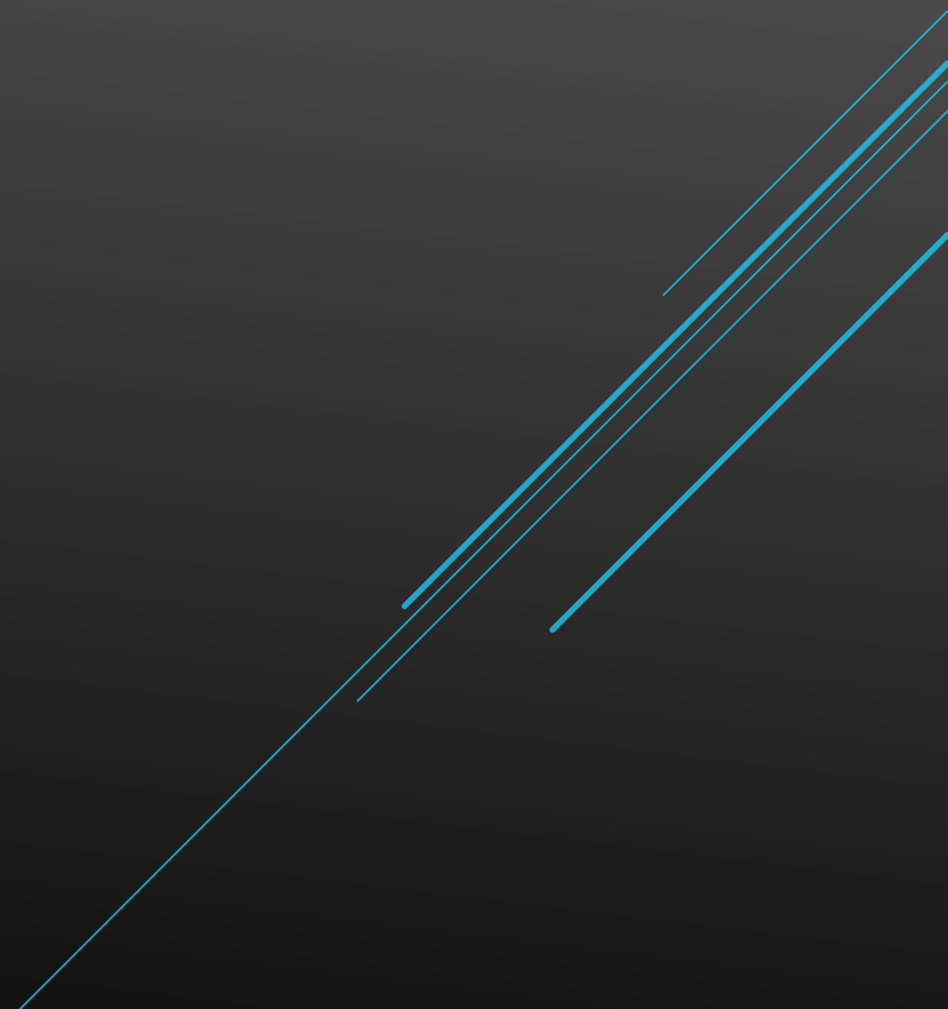


MS1411: MATHEMATICAL STATISTICS (WEEK 3)

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- ▶ Continuous Random Variables
 - ▶ Selected Continuous Distributions
 - ▶ Integration of Distribution Functions
 - ▶ Gamma & Chi-Squared Distributions
 - ▶ Normal Distribution & Standard Normal Conversion
 - ▶ Joint Distributions & Marginal Densities
 - ▶ Covariance & Correlation
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- ▶ Relevant Reading (Walpole et al. 9th ed.):
 - ▶ Continuous Random Variables: Chapter 3.3
 - ▶ Joint Distributions: Chapter 3.4
 - ▶ Covariance & Correlation: Chapters 4.1–4.4
 - ▶ Continuous Distributions: Chapters 6.1, 6.6, 6.7, 6.10

THIS WEEK'S FOCUS



► **Definition:**

- A random variable X is continuous if it can take on infinitely many values over some interval.

► **Probability Density Function (PDF), $f(x)$:**

- $P(a \leq X \leq b) = \int_a^b f(x)dx$
- $f(x) \geq 0$ for all x and $\int_{-\infty}^{\infty} f(x)dx = 1$

► **Cumulative Distribution Function (CDF), $F(x)$**

- $F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$

► **No Probability at a Single Point:**

- For continuous X , $P(X=c)=0$.

► **Why Continuous Models Matter**

- They describe real-world measures that vary smoothly (e.g., height, weight, time, voltage).
- They form the foundation for many important distributions (Normal, Exponential, etc.) used in engineering and science.

CONTINUOUS RANDOM VARIABLES — FOUNDATIONS

- ▶ We will focus on the following well-known distributions:
 1. **Uniform Distribution**
 2. **Gamma Distribution** (with special case the **Exponential Distribution**)
 3. **Chi-Squared Distribution**
 4. **Normal Distribution**
- ▶ **Why These Matter:**
 - They frequently arise in engineering & science contexts (e.g., waiting times → exponential, measurement errors → normal).
 - They form building blocks for more advanced models.
 - ▶ (Ref: Walpole Ch. 6.1, 6.6, 6.7, 6.10)

KEY CONTINUOUS DISTRIBUTIONS — OUTLINE

UNIFORM DISTRIBUTION

► **Definition:**

A continuous random variable X is said to follow a *Uniform* distribution on the interval $[a,b]$ if its probability density is *constant* across $[a,b]$. Formally,

$$X \sim \text{Uniform}(a,b) \Leftrightarrow f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

► **Interpretation**

- "All sub-intervals of $[a,b]$ are equally likely."
- If you pick a value randomly from $[a,b]$, every portion of equal length has the same probability.

► **Probability Calculations**

- Probability of x between c and d (where $a \leq c < d \leq b$)

$$P(c \leq X \leq d) = \int_c^d \frac{1}{b-a} dx = \frac{d-c}{b-a}.$$

- CDF $F(x)$ is:

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

UNIFORM DISTRIBUTION

► Mean and Variance

- **Mean:** $E[X] = \frac{a+b}{2}$.
- **Variance:** $Var(X) = \frac{(b-a)^2}{12}$.

► Interpretation

- The center of the interval $\frac{a+b}{2}$ is the “balance point.” The spread $\frac{(b-a)^2}{12}$ shows that variability grows with the square of the width $b-a$.

► Illustrative Example

- Suppose $X \sim \text{Uniform}(0, 10)$:
 - PDF: $f(x) = \frac{1}{10}$ for $0 \leq x \leq 10$
 - Mean = $\frac{0+10}{2} = 5$
 - Variance = $\frac{(10-0)^2}{12} = \frac{100}{12} = 8.33$
- Probability that X lies between 2 and 6:
 - $P(2 \leq X \leq 6) = \int_2^6 \frac{1}{10} dx = \frac{6-2}{10} = 0.4$

► Why Uniform Matters

- Often used to model a “lack of knowledge” or “all outcomes equally likely” scenario.
- Forms a building block in simulation methods (e.g., pseudo-random number generation).

- ▶ The Gamma distribution is a continuous family of distributions that extends the Exponential distribution and is widely used for modeling **waiting times, lifetimes, and processes with “gamma-shaped” failure rates.**
- ▶ **Gamma Function**
 - ▶ The gamma function $\Gamma(\alpha)$ is defined as:
 - ▶ $\Gamma(\alpha) = \int_0^{\infty} z^{\alpha-1} e^{-z} dz,$
 - ▶ And satisfies:
 - ▶ $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha).$
 - ▶ **Key Fact (Integers):** $\Gamma(n)=(n-1)!$ for any positive integer n .
 - ▶ So, for example, $\Gamma(4)=3! = 6.$
- ▶ **Gamma Distribution PDF**
 - ▶ Let X be a random variable that follows the Gamma distribution with parameters α (*shape*) and β (*scale*). We write $X \sim \Gamma(\alpha, \beta)$. The PDF is:
 - ▶ $f(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}}, x > 0, \alpha > 0, \beta > 0$
 - ▶ Interpretation (Shape & Scale):
 - ▶ α (shape) influences the “skewness” and peak.
 - ▶ β (scale) stretches or shrinks the distribution along the x -axis.

GAMMA DISTRIBUTION & ITS PROPERTIES

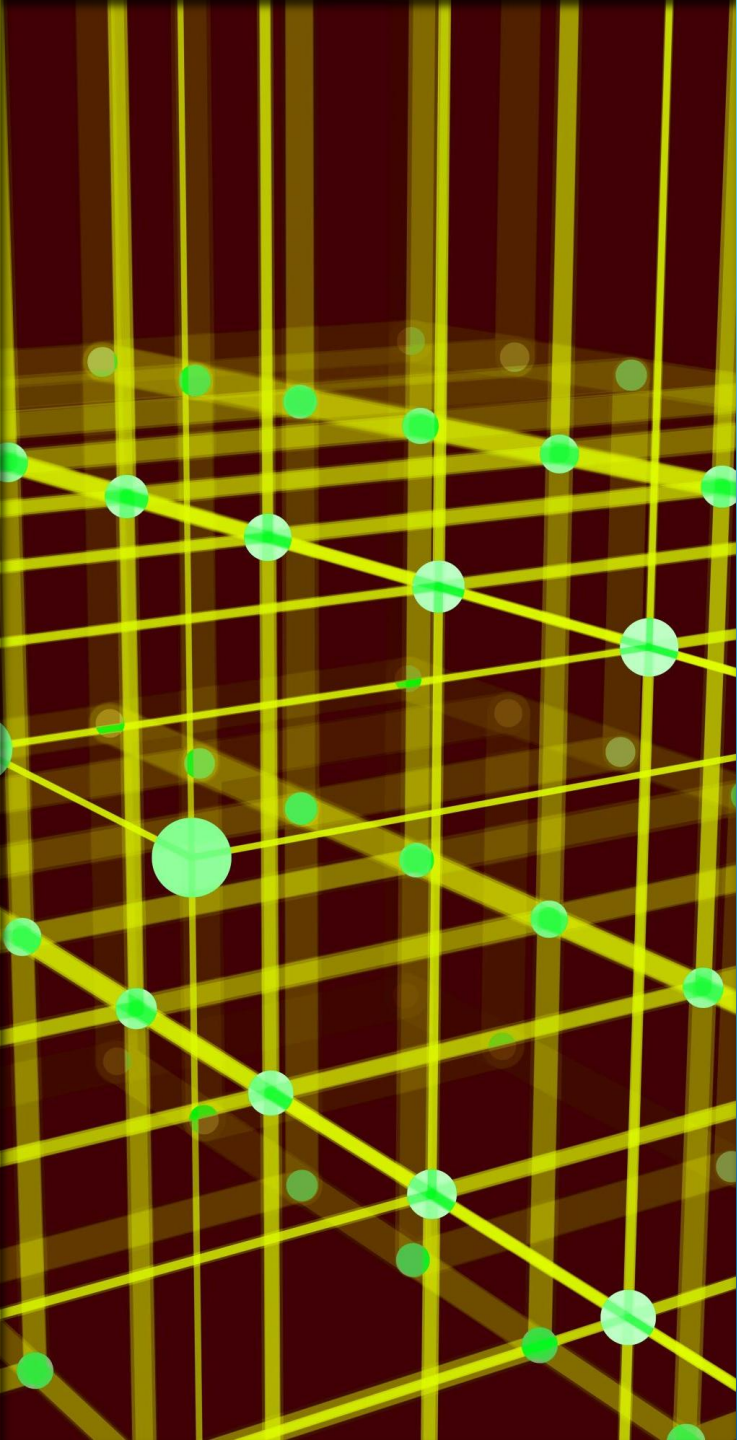
► **Mean & Variance**

- From the PDF, we can derive:
 - $E[X] = \alpha\beta$,
 - $Var(X) = \alpha\beta^2$
 - Larger α shifts the center to the right and can reduce skewness.
 - Larger β widens the spread of the data.

► **Exponential Distribution (Special Case)**

- When $\alpha=1$, the Gamma distribution becomes the Exponential distribution:
- $f(x; \beta) = \frac{1}{\beta} e^{-\frac{x}{\beta}}, x > 0$
- Equivalently, Exponential $\sim \Gamma(1, \beta)$.
- Why It Matters:
 - The exponential is used to model the waiting time until the first event in a Poisson process.
 - The gamma distribution models the waiting time until the α -th event in a Poisson process (when α is an integer).

GAMMA DISTRIBUTION & ITS PROPERTIES

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- ▶ The **Chi-Squared** distribution (χ^2) is a special case of the Gamma distribution, crucial in statistics for analyzing **population variance**, **goodness-of-fit tests**, and **many other inference procedures**.
 - ▶ **Definition:**
 - ▶ A random variable X follows a chi-squared distribution with r degrees of freedom if:
 - ▶ $X \sim \chi_r^2 \Leftrightarrow X \sim \Gamma\left(\frac{r}{2}, 2\right)$.
 - ▶ Equivalently, the PDF is:
 - ▶ $f(x; r) = \frac{1}{2^{\frac{r}{2}} \Gamma(\frac{r}{2})} x^{\frac{r}{2}-1} e^{-\frac{x}{2}}, x > 0$
 - ▶ **Degrees of Freedom (r):**
Represents the number of independent “pieces of information” in many sampling or test settings (e.g., sum of squares of sample data).
 - ▶ **Mean & Variance**
 - ▶ $E[X] = r, \text{Var}(X) = 2r$
 - ▶ Intuition:
 - ▶ The center of the distribution is at r .
 - ▶ The variability grows proportionally to r .

CHI-SQUARED DISTRIBUTION



► **Origin from Normal Distributions:**

► If $Z_1, Z_2, \dots, Z_r$ are **independent $N(0,1)$** random variables, then:

►
$$X = Z_1^2 + Z_2^2 + \dots + Z_r^2$$

► Follows a χ^2_r distribution.

► This is central in **analysis of variance, confidence intervals for σ^2** , and many **test statistics**.

► **Using χ^2 Tables**

► Tables generally show critical values x_α for upper-tail probabilities $P(X > x_\alpha)$.

► For example, if $X \sim \chi^2_{10}$, you might check the table row for $r=10$ to find the critical value corresponding to a 5% tail ($\alpha=0.05$).

► **Common Applications**

► **Testing Population Variance:**

► Hypothesis tests about σ^2 , e.g., $\chi^2 = \frac{(n-1)S^2}{\sigma^2}$.

► **Goodness-of-Fit Test:**

► Check if observed data match a proposed distribution.

► **Independence Tests in Contingency Tables:**

► χ^2 -based tests for categorical data (e.g., independence or homogeneity).

CHI-SQUARED DISTRIBUTION

- ▶ The **Normal (Gaussian)** distribution is by far the most widely used continuous distribution, crucial in measurement error analysis, natural phenomena modeling, and large-sample theory (Central Limit Theorem).

- ▶ **Definition & PDF**

- ▶ A random variable X is **normally distributed** with mean μ and variance σ^2 if

- ▶ $X \sim N(\mu, \sigma^2) \Rightarrow f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty.$

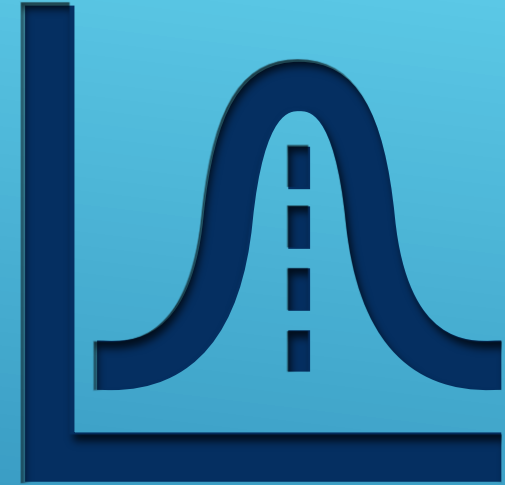
- ▶ **Shape:** Bell-shaped, symmetric about μ .

- ▶ Parameters:

- ▶ μ determines the center.
 - ▶ σ controls spread (standard deviation).

- ▶ **Empirical (68–95–99.7) Rule**

- ▶ For $X \sim N(\mu, \sigma^2)$:
 - ▶ ~68% of observations lie within 1 standard deviation of μ . $P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$
 - ▶ ~95% lie within 2 standard deviations. $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$
 - ▶ ~99.7% lie within 3 standard deviations. $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$



NORMAL DISTRIBUTION — KEY INSIGHTS

- ▶ **Standard Normal Distribution**

- ▶ **Key Transformation:**

- ▶ $Z = \frac{X - \mu}{\sigma} \sim N(0,1)$

- ▶ **Table Lookup:**

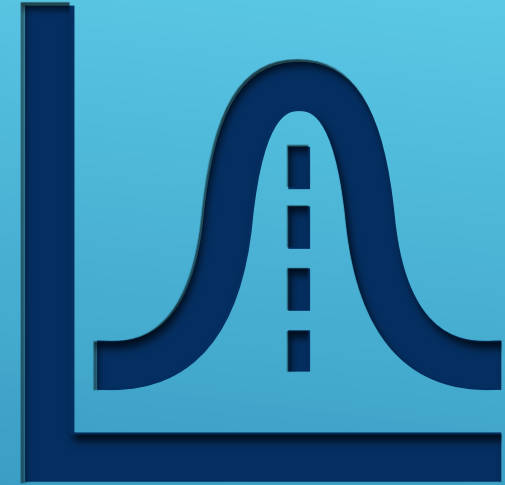
- ▶ Probability statements about X become statements about Z .
 - ▶ Statistical software or standard normal tables give $P(Z \leq z)$.

- ▶ **Example:**

- ▶ $P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)$.

- ▶ **Why the Normal Distribution Is Everywhere**

- ▶ **Measurement Errors** often sum many small effects, yielding a bell-shaped distribution.
 - ▶ **Central Limit Theorem (CLT)** states that averages of large samples approach a normal distribution, no matter the original population's shape.
 - ▶ Useful in **quality control**, **regression analysis**, **confidence intervals**, and **hypothesis testing**.



NORMAL DISTRIBUTION — KEY INSIGHTS

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- ▶ For $X \sim N(\mu, \sigma^2)$, we define
 - ▶ $Z = \frac{X - \mu}{\sigma}$.
 - ▶ Then $Z \sim N(0, 1)$. Probabilities involving X can be transformed via Z-scores:
 - ▶ $P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)$.
 - ▶ Table Lookup Example:
 - ▶ If $Z = 1.28$, from Normal Table, $P(Z \leq 1.28) \approx 0.8997$.

CONVERTING TO THE STANDARD NORMAL

JOINT DISTRIBUTIONS (TWO OR MORE VARIABLES)

- ▶ When **two (or more) continuous random variables** are observed together, we need a function that describes their behavior simultaneously; this is called the **joint probability density function** (joint PDF).
- ▶ **Joint PDF for Continuous Variables**
 - ▶ For two continuous random variables X and Y , the joint PDF $f_{X,Y}(x, y)$ satisfies:
 - ▶ $f_{X,Y}(x, y) \geq 0$ for all (x, y)
 - ▶ $\iint_{\mathbb{R}^2} f_{X,Y}(x, y) dx dy$
 - ▶ Probability over a region:
 - ▶ $P((X, Y) \in A) = \iint_{A^2} f_{X,Y}(x, y) dx dy,$
 - ▶ where A is a region in the xy -plane.
- ▶ **Marginal Distributions**
 - ▶ To get individual information about X or Y , we “integrate out” the other variable:
 - ▶ Marginal of X : $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy,$
 - ▶ Marginal of Y : $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx,$
 - ▶ Meaning: $f_X(x)$ is the PDF of X alone, found by summing (integrating) over all possible values of Y .

JOINT DISTRIBUTIONS (TWO OR MORE VARIABLES)

- ▶ Independence
 - ▶ X and Y are independent if their joint PDF factors into the product of their marginals:
 - ▶ $f_{X,Y}(x,y) = f_X(x)f_Y(y)$.
 - ▶ Interpretation: Knowing $Y=y$ tells us nothing extra about X ; their behaviors are “decoupled.”
- ▶ Conditional PDFs
 - ▶ Even if X and Y are not independent, we can define the conditional PDF:
 - ▶ $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$, provided $f_Y(y) \neq 0$.
 - ▶ This describes the distribution of X given a specific value $Y=y$.
- ▶ Practical Importance
 - ▶ Used in joint modeling of multiple responses (e.g., height and weight, or stress and strain).
 - ▶ Forms the basis for understanding covariance and correlation between variables (see next slides).
 - ▶ Integral in multivariate statistical analysis, where dimensionality can be greater than 2.

- ▶ When dealing with two random variables X and Y , we often want to measure **how and how strongly** they vary together. Two key measures are **covariance and correlation**.
- ▶ **Covariance**
 - ▶ Definition:
 - ▶ **Covariance**: This tells us how two variables “move” together (linearly). If two variables tend to increase or decrease at the same time, they have a **positive** covariance. If one tends to go up when the other goes down, their covariance is **negative**. If there is no consistent upward or downward link, their covariance will be **close to zero**. However, because covariance can be any positive or negative number (including large magnitudes), it depends on the actual measurement units of the variables.
 - ▶ $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y]$
 - ▶ Interpretation of Sign:
 - ▶ $Cov(X, Y) > 0$: X and Y **increase together** (positive association).
 - ▶ $Cov(X, Y) < 0$: X and Y **move in opposite directions** (negative association).
 - ▶ $Cov(X, Y) = 0$: No linear relationship (or purely no “average” linear trend) – but not necessarily independence.
 - ▶ Units:
 - ▶ The unit of $Cov(X, Y)$ is the product of the units of X and Y .
 - ▶ Example: If X is in cm and Y is in grams, $Cov(X, Y)$ is in cm · grams.

COVARIANCE & CORRELATION

► Correlation Coefficient

► Definition:

- **Correlation:** This is essentially a “scaled” version of covariance, adjusted so that its values always lie between -1 and $+1$. A correlation near $+1$ means the two variables have a **strong positive** linear relationship (they rise and fall together). A correlation near -1 means a **strong negative** linear relationship (one goes up while the other goes down). A correlation near 0 means **no strong linear** relationship. Correlation is unitless, so it is often more interpretable than raw covariance.

- $$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1,1].$$

► Interpretation of Magnitude:

- $|\rho| \approx 1$: Strong linear association (perfect linear relation if $\rho = \pm 1$).
- $|\rho| \approx 0$: Weak or no linear association.

► Interpretation of Sign:

- $\rho > 0$: X and Y tend to increase together.
- $\rho < 0$: X and Y tend to move oppositely.

- **Unitless:** Correlation is a pure number, free of measurement units, allowing comparisons across different data sets.

COVARIANCE & CORRELATION

► Why Covariance & Correlation Matter

► Assessing Relationships:

- In engineering, we may examine how temperature and pressure fluctuate together.
- In biomedical studies, we check how cholesterol level correlates with blood pressure.

► Regression & Prediction:

- High correlation suggests a strong linear model might be effective.
- Low correlation implies smaller linear predictive power (though nonlinear patterns could still exist).

► Statistical Independence:

- $\text{Cov}(X,Y)=0$ or $\rho=0$ does not guarantee independence—only no linear trend.
- Independence implies $\text{Cov}(X,Y)=0$, but the converse can fail when relationships are nonlinear.

► Simple Example

- Let (X_i, Y_i) for $i=1, \dots, n$ be sample data. Then the **sample covariance** is

- $$s_{XY} = \frac{1}{1-n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}),$$

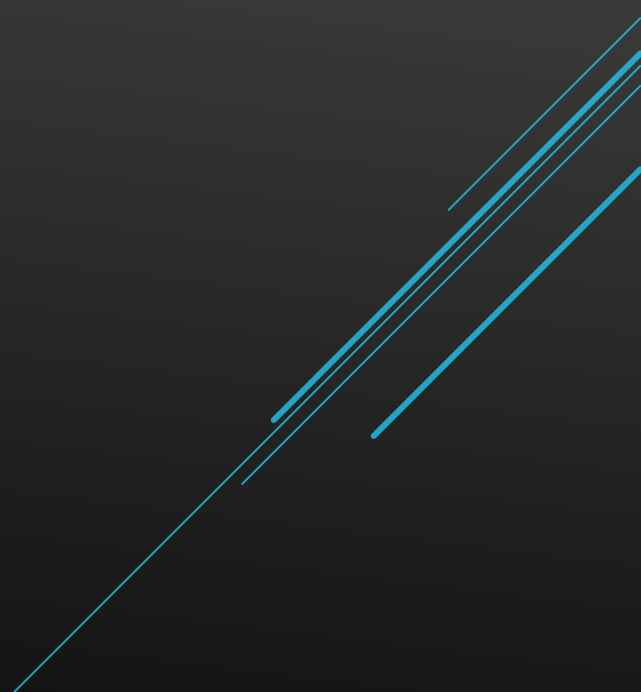
- and the **sample correlation** is

- $$r = \frac{s_{XY}}{\sqrt{s_X^2 s_Y^2}},$$

- where s_X^2 and s_Y^2 are the sample variances of X and Y .

COVARIANCE & CORRELATION

SUMMARY & TIPS

- ▶ Continuous Variables revolve around integrals of PDFs and CDFs.
 - ▶ Gamma & Chi-Squared Distributions appear in reliability analyses and variance-based tests.
 - ▶ Normal Distribution is central in many theoretical and practical contexts (error analysis, CLT).
 - ▶ Standard Normal Table usage is a key skill.
 - ▶ Joint Distributions describe pairs (X,Y) , from which marginals are derived.
 - ▶ Covariance & Correlation measure linear association between random variables.
- 
- A series of parallel teal lines of varying lengths and orientations, located in the bottom right corner of the slide, creating a modern, abstract graphic element.

- ▶ **Part A:** Let X be a continuous random variable uniformly distributed on the interval $[2,6]$.
 - ▶ Write down the probability density function (PDF) $f_X(x)$.
 - ▶ Find $P(3 \leq X \leq 4)$.
 - ▶ Compute the mean μ_X and variance σ_X^2 of X .
- ▶ **Part B:** Let $Y \sim N(\mu, \sigma^2)$ with $\mu=100$ and $\sigma=2$.
 - ▶ Convert $P(Y \leq 13)$ into a standard normal probability $P(Z \leq z)$.
 - ▶ Look up (or approximate) $P(Z \leq z)$ in the standard normal table (or using software).
 - ▶ Interpret your answer in the context of the distribution of Y .

EXERCISE 1

- ▶ **Part A (Gamma/Exponential):** Suppose X follows an exponential distribution with parameter $\beta=10$. That is, $X \sim \Gamma(1, 10)$.
 - ▶ Write down the PDF $f_X(x)$.
 - ▶ Compute $P(X \leq 5)$.
 - ▶ Find the mean and variance of X .
 - ▶ If Y is the waiting time until the second event in the same process, what Gamma distribution would Y follow? (Hint: the Gamma distribution is the sum of multiple exponentials.)
- ▶ **Part B (Chi-Squared):** Let $W \sim \chi_r^2$ with $r=6$ degrees of freedom.
 - ▶ State the PDF of W in terms of the gamma function.
 - ▶ Use a chi-squared table (or software) to approximate $P(W > 10)$.
 - ▶ Identify the table value or the method used.
 - ▶ Briefly explain how the chi-squared distribution arises from sums of squared standard normal variables.

EXERCISE 2